

# Learning language-model RL from an official reproduction

We want to establish that our language-model reinforcement learning pipeline does what we think it does: generate answers, assign rewards, change weights, and evaluate those changed weights fairly. We start with the authors’ code for *Understanding R1-Zero-Like Training: A Critical Perspective* [1,2]. This is a small reproduction of its training method, with a smaller model workload and budget. It does not reproduce the paper’s published scores.

## Why do this before recursive language-model RL?

In this experiment, the language model receives a mathematics question and writes one response. A checker compares its final answer with a reference. There are no live Python tools or recursive model calls in these responses. That makes the reward and update easier to inspect.

An RLM adds choices about decomposition, running code, reading intermediate results, and calling other models. A failure can then come from the language model, the surrounding software, the reward, or which actions receive credit. A working ordinary LLM experiment gives us a reference for the shared training machinery. It does not prove that those additional RLM decisions are trained correctly. We will still need separate checks of action boundaries, tool results, costs, and credit assignment.

## Pretraining, SFT, and RL are different sources of learning

| Stage                        | What supplies the learning target?                                                                          |
|------------------------------|-------------------------------------------------------------------------------------------------------------|
| Pretraining                  | Predicting text from a large existing collection develops the starting language and mathematical abilities. |
| Supervised fine-tuning (SFT) | Supplied example responses tell the model which tokens to imitate for a prompt.                             |
| Reinforcement learning (RL)  | The model samples its own responses. Rewards change how likely those responses are under its policy.        |

Our starting checkpoint is the **Qwen2.5-Math-1.5B base model** [3]. We add no SFT stage. “Base” does not mean untrained: its weights already contain substantial capabilities from earlier training. Calling this R1-Zero-like training does not mean beginning with random weights.

## What has been established so far?

Training with the chat prompt improved its full-500 score from **154 to 308**. But the base model already scored **305 with the question alone**. On question-only evaluation, the chat-trained model scored **317** and a separate question-trained model scored **314**. These modest gains over the stronger baseline need replication. Pages 4–5 distinguish training format, evaluation format, and completion behavior. Page 6 uses the smaller pilot to illustrate what an individual score change can mean.

Evidence cutoff: 7 October 2026, 11:34 UTC. Two training runs evaluated in both formats. Repeat status updated at 11:42 UTC.

## How eight answers produce an update

For each training question, the generator samples eight responses at temperature 1. The author reward function extracts the last boxed answer and checks it against the reference. A correct answer earns 1; a wrong or unparseable answer earns 0. There is no separate reward just for using a box. The training actor also assigns 0 to a response stopped by the token limit. The checker evaluates the answer, not the truth of every reasoning step [2].

### A teaching example: two successes in a group of eight

Suppose one question produces rewards

$$r = (1, 0, 0, 1, 0, 0, 0, 0), \quad \bar{r} = 2/8 = 0.25.$$

Dr. GRPO subtracts this group’s mean from each reward:

$$A_i = r_i - \bar{r}, \quad A = (0.75, -0.25, -0.25, 0.75, -0.25, -0.25, -0.25, -0.25).$$

The two successful responses receive positive advantages; the other six receive negative advantages. The update pushes toward the successful sampled responses and away from the unsuccessful ones. If all eight answers are wrong, or all eight are correct, their advantages are all zero. The sum being zero in a mixed group does not cancel the learning signal: different responses have different token probabilities and gradients.

### What is special about Dr. GRPO?

Dr. GRPO is the authors’ variant of Group Relative Policy Optimization. It does not divide advantages by the group’s reward standard deviation. It also sums token losses and divides by a constant response limit, here 3000, instead of dividing each response’s loss by its own length. These are the method’s two normalization changes [1,2]. Its policy loss uses a clipped probability ratio:

$$q_{it} = \frac{\pi_{\theta}(y_{it} \mid x_i, y_{i,<t})}{\pi_{\text{old}}(y_{it} \mid x_i, y_{i,<t})}, \quad L = -\frac{1}{B \, 3000} \sum_{i=1}^B \sum_{t \in \text{response}_i} \min(q_{it} A_i, \text{clip}(q_{it}, 0.8, 1.2) A_i).$$

Here the policy is the model’s next-token distribution,  $\theta$  is its set of weights, and  $B = 128$  responses. The group advantage is shared across all response tokens; this experiment does not determine which individual reasoning step deserves credit. We use zero KL penalty: no extra penalty keeps the updated policy close to a reference model.

### How the single-GPU run fits together

One collection contains 16 questions times eight responses: 128 trajectories. The learner processes one trajectory at a time, accumulates 128 gradients, and takes one optimizer update. It then sends the new weights to vLLM for the next collection. We use one pass over each collection, full-model BF16 training, learning rate  $10^{-6}$ , and the author’s Adam settings.

The pilot’s 64 questions gave four updates; each completed 512-question run gave 32 updates from 4096 sampled responses. Training and generation alternate on one A100 with 40 GB of memory. Putting the generator to sleep frees memory; it does not reset the weights. Likewise, switching a model between training and evaluation modes changes its execution behavior, not its learned weights. The optimizer performs the update.

## Freeze the questions and the evaluator

We use the official code at commit `dfca49dd460e`, Oat 0.1.3.post1, vLLM 0.8.4, and Qwen model revision `4a83ca6e4526`. Full identifiers, package versions, and content hashes are recorded beside this guide in the research record and data manifest [4].

| Purpose          | Size | Selection and use                                                                                                                  |
|------------------|------|------------------------------------------------------------------------------------------------------------------------------------|
| Pilot training   | 64   | First 64 indices after shuffling 12,000 author training rows with Python <code>random.Random(42)</code> .                          |
| Larger training  | 512  | First 512 indices from the same shuffle. This includes the pilot questions; both prompt-format runs restart from the base weights. |
| Monitoring       | 64   | Shuffle the author’s 500 MATH evaluation rows with seed 142; take the first 64. Reused for progress checks.                        |
| Final comparison | 128  | Take the next 128 from that evaluation shuffle. Disjoint from monitoring; used for the larger run’s fixed final comparison.        |

The 12,000 training rows include 4,501 IDs from the original MATH test split. An arbitrary download of that test split would therefore be a poor choice of evaluation data. We checked the author’s supplied training collection against the supplied MATH500 questions and found no exact overlaps after whitespace normalization. This does not rule out paraphrases or exposure during the base model’s pretraining.

### Keep three kinds of settings separate

**Weights.** The baseline uses the original base checkpoint. Each trained model uses its prescribed final checkpoint. The two 512-question runs start separately from the base, not from the pilot or each other’s weights.

**Generation.** Each before/after pair uses the same input format, the same 3000-token response limit, and greedy decoding with one answer per question. We evaluate both the Qwen-Math chat prompt and the question alone; page 5 shows what changes. Training uses temperature 1 for varied samples. Comparing base and trained weights under different prompts would mix prompt effects with weight changes. This is a central warning of the paper [1].

**Scoring.** The author’s training program uses a faster checker. The standalone evaluator uses an additional mathematical-equivalence check (`fast=False`). We apply the same standalone checker to both saved answer sets for every reported comparison, including both formats for the newly completed question-trained model. Source questions and reference answers were checked against the frozen data. We never combine scores from different checkers.

### Choose the checkpoint before reading the final score

We retain the prescribed final checkpoint and evaluation, even if an earlier evaluation looks better. On the 64 monitoring questions, the chat-trained run’s final evaluation scored 36, versus 20 at baseline; another evaluation after the same 32 updates scored 39. The pilot likewise scored 23 and then 22 with identical saved weights. Greedy GPU decoding was not exactly repeatable. We report this variation without assigning it a specific numerical cause or choosing the higher score.

Learner seed 42 is fixed, but the unchanged author actor chooses a time-based generation seed. The run is therefore not fully deterministic. The smaller question-only gain especially needs evaluation repeats and independent training runs to assess its reproducibility. A fixed seed label alone does not supply them.

### 3 / TWO COMPLETED TRAINING RUNS

## Training format and evaluation format both matter

Two separate runs restarted from the original Qwen2.5-Math-1.5B base weights. Each used the same 512 questions, learner seed 42, 4096 sampled responses, and 32 full-model BF16 updates. One trained with the chat prompt; the other trained with the question alone. Both used the unchanged author algorithm. We compare their prescribed final checkpoints with the starting weights on all 500 questions, using the full checker [4].

| Evaluation input      | Starting weights | Trained with chat prompt | Trained with question only |
|-----------------------|------------------|--------------------------|----------------------------|
| Qwen-Math chat prompt | 154/500          | 308/500                  | 168/500                    |
| Question alone        | 305/500          | 317/500                  | 314/500                    |

Within each row, models are compared under matched evaluation settings. Between rows, the input format changes. The two trained columns come from different training runs. The question-only training follow-up was chosen after seeing the stronger question-only baseline; it is exploratory.

On question-only evaluation, chat-prompt training gains 12 answers (18 gains, six losses), while question-only training gains nine (24 gains, 15 losses). These are increases of 2.4 and 1.8 percentage points over 61.0%. The three-answer difference between trained models does not establish which training format is more reliable. With the chat evaluation prompt, question-only training changes 154 to 168, with 21 gains and seven losses.

### What do the different gains mean?

The large gain remains the chat-trained model under chat evaluation, from 30.8% to 61.6%. Gains against the strong question-only baseline are modest in both runs. These observations do not establish newly acquired mathematics, zero learning, or a single cause for every changed answer. They also do not validate RLM tool use or credit assignment across multiple calls.

Chat evaluation had 205 capped responses initially, 29 after chat-prompt training, and 201 after question-only training. Question-only evaluation had 15, 17, and 19 respectively. Its mean response lengths were 694.31, 691.10, and 693.61 tokens. The new question-trained model’s chat responses averaged 1616.55 tokens, compared with 1628.04 initially. These are descriptive completion measurements; all new final-evaluation responses were nonempty.

### Keep the original fixed test intact

For the chat-trained model, the separate prescribed 128-question comparison remains **41/128 before and 80/128 after**, with 44 gains and five losses. The newly generated full-500 batch contains those same 128 questions, but its chat-format counts are 41 and 82. That later observation does not replace the original result. Changing evaluation batches can change generated answers.

In the original comparison, capped responses fell from 56 to 12. Of its 44 gains, 36 had a capped baseline response and 37 had no parseable baseline answer. Those categories overlap. They identify completion as an important limitation, but cannot distinguish changes in reasoning, repetition, and answer formatting on their own.

## A chat prompt is extra input, not new weights

The official `qwen_math` option wraps each question in these role markers and adds a system instruction. The placeholder below stands for the unchanged dataset question [2].

```
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{}.<|im_end|>
<|im_start|>user
[question]<|im_end|>
<|im_start|>assistant
```

The official `no` option passes just the question text. It adds no role markers and no instruction about reasoning or boxing the answer. Both formats still use the same model architecture, greedy decoding, 3000-token response limit, and final-answer checker. The question-only baseline demonstrates that this pretrained base model can already answer many questions without those extra instructions. Only the base model’s input format has changed.

### Which questions make up the full-500 result?

All 500 questions were excluded from our weight training. Their roles in this study nevertheless differ: 64 were repeatedly monitored, 128 formed the original fixed test, and 308 add further coverage. These counts come from the fresh full-500 evaluations, with both trained checkpoints kept separate:

| Input         | Question role     | Starting | Trained<br>with chat | Trained<br>with questions |
|---------------|-------------------|----------|----------------------|---------------------------|
| Question only | Monitored 64      | 39       | 37                   | 39                        |
|               | Original test 128 | 79       | 84                   | 84                        |
|               | Additional 308    | 187      | 196                  | 191                       |
| Chat prompt   | Monitored 64      | 21       | 42                   | 24                        |
|               | Original test 128 | 41       | 82                   | 45                        |
|               | Additional 308    | 92       | 184                  | 99                        |

The additional questions broaden the evaluation, but the full set is not 500 entirely uninspected questions. More evaluation rows are not more training replicates. We do not know whether these problems appeared during the base model’s pretraining.

### Check the actual inputs, then repeat the result

The 10:35 question-only attempt was rejected: despite its command-line setting, a dataset cache supplied chat-formatted training inputs. The valid fresh run used an isolated cache, started at 10:40, and finished by 11:24. All 4096 saved training prompts were checked against the 512 raw source questions. A setting in a configuration file is not evidence of what the model actually received.

The unchanged-weight evaluations repeated exactly: the same saved answers and scores of 305 and 314. A fresh question-only training run with learner seed 43 started at 11:41; its result is pending. We retain all results, not just the highest score. Re-evaluating one saved model measures variation in generation; training again measures variation in the learning process. Neither check should be replaced by choosing a favorable checkpoint or answer batch.

## What did the pilot’s two extra points mean?

Under the standalone checker, 19 monitoring answers stayed correct, 41 stayed incorrect, three became correct, and one became incorrect. Thus the net change is  $+2/64$ , or 3.125 percentage points. These are the earlier pilot’s **monitoring results**, separate from the larger run’s 41-to-80 test result.

| Changed problem | What the saved responses show                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------------------------------------|
| Fuel use        | Repetition without an answer became a completed calculation and boxed 550.                                                |
| Magic square    | The baseline already derived $n = 7$ , then repeated an incorrect verification. The final response completed and boxed 7. |
| Recurrence      | A boxed answer changed from 1089 to the correct 898. Both responses had valid answer formatting.                          |
| Trigonometry    | A correct boxed 12 became repetitive noncompletion at the 3000-token limit.                                               |

### A concrete example: getting an answer out reliably

Karla drove 12,000 miles. How much fuel would she save with a car averaging 48 miles per gallon instead of 15? The completed post-training response gives

$$\frac{12000}{15} = 800, \quad \frac{12000}{48} = 250, \quad 800 - 250 = \boxed{550}.$$

The baseline repeated a version of the question until the 3000-token limit. The final response completed in approximately 417 text tokens. This is useful behavioral improvement on that example, but it does not show that the model newly acquired division or subtraction.

The recurrence example is also instructive. Independent calculation gives  $211 + 420 + 267 = 898$ . Both responses wrote Python describing the correct recurrence, but the baseline’s claimed code output was 1089. Those fenced “output” blocks were generated text; no Python tool had run. A correct final answer does not validate an entire explanation or prove successful tool use.

### What the training evidence adds

The first collection had 12 rewarded responses out of 128, with seven mixed groups giving 56 nonzero advantages. Twenty responses reached the token limit. The four completed optimizer updates had finite nonzero gradients, sampled checkpoint tensors changed, and updated weights were sent back to the generator. This establishes an operational path from sampled answers to weight updates. It does not make the small monitoring gain statistically conclusive.

The pilot illustrates why an aggregate score needs accompanying examples. The larger chat-format gain and the modest question-only gains both need the same care in interpretation.

#### Sources and editable evidence.

[1] Liu et al., Understanding R1-Zero-Like Training: A Critical Perspective. <https://arxiv.org/abs/2503.20783>.

[2] Pinned official training, evaluation, and grading code; Oat package version.

[3] Pinned Qwen2.5-Math-1.5B model revision.

[4] Companion [research record](#), [data manifest](#), and [resolved environment](#); [main training receipt](#); [fixed-128 scoring receipt](#); [full-500 chat-format receipt](#); [prompt-comparison receipt](#); [question-only training receipt](#); [question-trained final scores](#); [pilot receipt](#). The research record links the question-only answer files and scoring evidence.